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A Case History Review for a High-Tier Fracturing Services Start-Up in KSA

A. Yudin, N. Nurlybayev, A. AlYaseen, Z. AlJalal, C. McCollum, and H. Yusuf, TAQA, Industrialization and energy services company, KSA

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Abstract

Oil & gas sector in the Middle East is rapidly evolving and local service providers start to play an increasingly bigger role. In the current study, the paper will capture key factors contributing to the steady development of the Saudi-based national fracturing company start-up to reach the level of a hi-tier fracturing solutions provider, both acid and proppant fracturing, within less than one year period.

Establishing the solid performance was based on identifying the main pillars to operate with high service quality and efficiency. Integrity and transparency of the quality management system were among the key elements to achieve continuous improvement during operation expansion. Highly experienced personnel to execute treatments and perform equipment maintenance were also cornerstones for reliable service delivery. Furthermore, access to fluid technologies and a strong team of stimulation experts allowed exponential growth for the start-up company to become an international service provider. Several hi-tier case histories were selected for this study to detail the approach and execution workflows.

Upon evaluating key metrics influencing the performance efficiency including, non-productive time (NPT), lost time incident (LTI), treatment pumping success ratio and initial flow rate, it was observed that the performance efficiency of the start-up company did not drop below 97%. Additionally, several hi-tear fracturing campaigns were successfully executed in various reservoir environments and conditions. For clastic rocks, a novel degradable diverting material, tailored to middle eastern environmental and geological conditions was developed and tested. For the carbonate rocks, several solutions were also locally tailored with improved performance such as extremely high-temperature reservoirs were treated with a combination of retarded acid system and viscoelastic surfactant-based including far-field diverting technology, that were alternated with viscous pills and inhibited hydrochloric acid. For the tight sectors of the reservoir, where fracturing fluid recovery was challenging, two foam-based fracturing systems were implemented delivering up to 40% faster fluid recovery and more than 15% higher initial well productivity. It was noted that CO2-based foam fracturing was adopted to be the standard fracturing technique for the start-up company within less than 12 months of operations.

Localization of services in the energy sector is expected to continue as a trend in the Middle East and other regions, so this comprehensive study of the fracturing national start-up with operations and quality

improvement workflows and technological case histories will be a valuable support for the engineers and managers to succeed in a similar type of projects.

Introduction to Geological and Operational Challenges

Saudi Arabia's conventional reservoirs present unique challenges related to reservoir heterogeneity, stress conditions, and stress complexity, as highlighted in multiple studies (Khan et al. 2020, Yudin et al. 2023, Nurlybayev et al. 2024, Yudin and Rivas et al. 2024, Yudin and Farouk et al. 2024). Addressing these challenges requires a comprehensive understanding of reservoir properties, precise engineering, and adaptive stimulation techniques. Sandstone reservoirs exhibit significant heterogeneity. Variations in rock properties, porosity, and permeability can lead to uneven fracture propagation during hydraulic fracturing treatments. The challenge lies in creating optimal hydraulic fractures that effectively connect the wellbore to the reservoir matrix. Uneven fracture distribution can impact production rates and ultimate recovery. These reservoirs often experience high stress and pressure. Reservoir pressures can reach up to 11,000 psi, and stress gradients can be as high as 1.1 psi/ft. Managing these extreme conditions during hydraulic fracturing requires precise engineering and robust design strategies. Achieving adequate fracture half-length and vertical coverage is crucial. Inadequate fracture half-length can limit connectivity to the reservoir, affecting post-fracture productivity. Designing treatments that address these limitations is essential for sustained gas production. Poor fracture fluid cleanup can hinder well productivity. Inadequate removal of proppant and other debris from the fracture can lead to reduced conductivity and suboptimal productivity. Strategies to enhance fluid cleanup are critical for maintaining optimal performance.

A tailored approach using various fluid compositions is common in the area. Reservoirs below 300°F are treated with borate-based crosslinkers, while those above 300°F typically use zirconate-based crosslinkers (Nurlybayev et al. 2023). Fracturing projects adopt an integrated approach, where fluid modeling and optimization are continuously improved. Reliable simulators are crucial in these projects to optimize team performance, avoid delays and non-productive time (NPT), and enhance well productivity.

The focus of our study is on conventional, tight, and deep gas reservoirs where operations are conducted in an integrated workflow. This approach involves various service product lines working together in a coordinated sequence to achieve optimal well completion, fracturing, and production. Any non-productive time (NPT) leads to significant losses due to equipment and service crews being on standby, delayed well production, and increased overall health, safety, and environmental (HSE) risks, which are top priorities for operators and regulatory bodies.

Historically, the industry separated technical decisions from operational and economic considerations. However, given the current trend towards conveyor-based operations where efficiency drives economic field development, technical experts must integrate various components of the problem to deliver an optimized technical solution. Fig 1 and Fig. 2 demonstrate reservoir completion and surface equipment layout respectively to give better visual representation of an integrated fracturing project complexity. Developing tight reservoir with marginal economics, it is important to optimize costs and logistics for the integrated fracturing projects. A more detailed view on this aspect of the on-going fracturing campaign was shared by Badurayq et al. 2024.

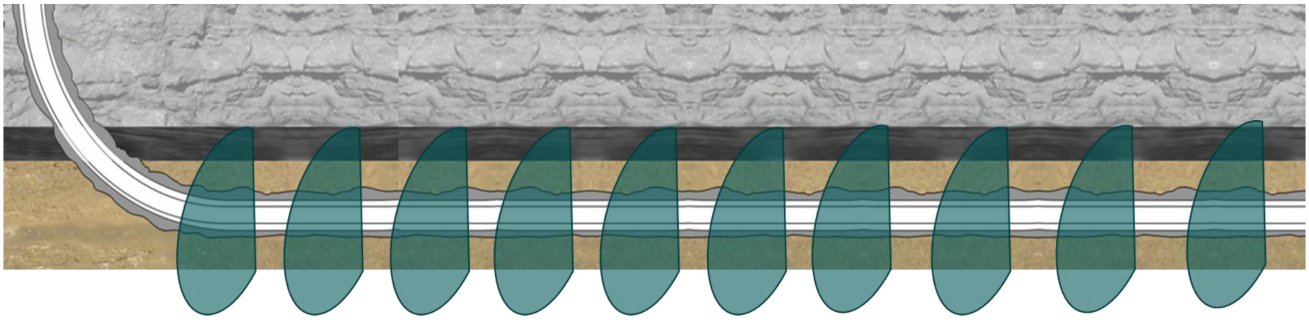


Figure 1—Schematical representation of 10-stages completion with fracturing in HPHT reservoir. (P&P cemented wellbore in this case).

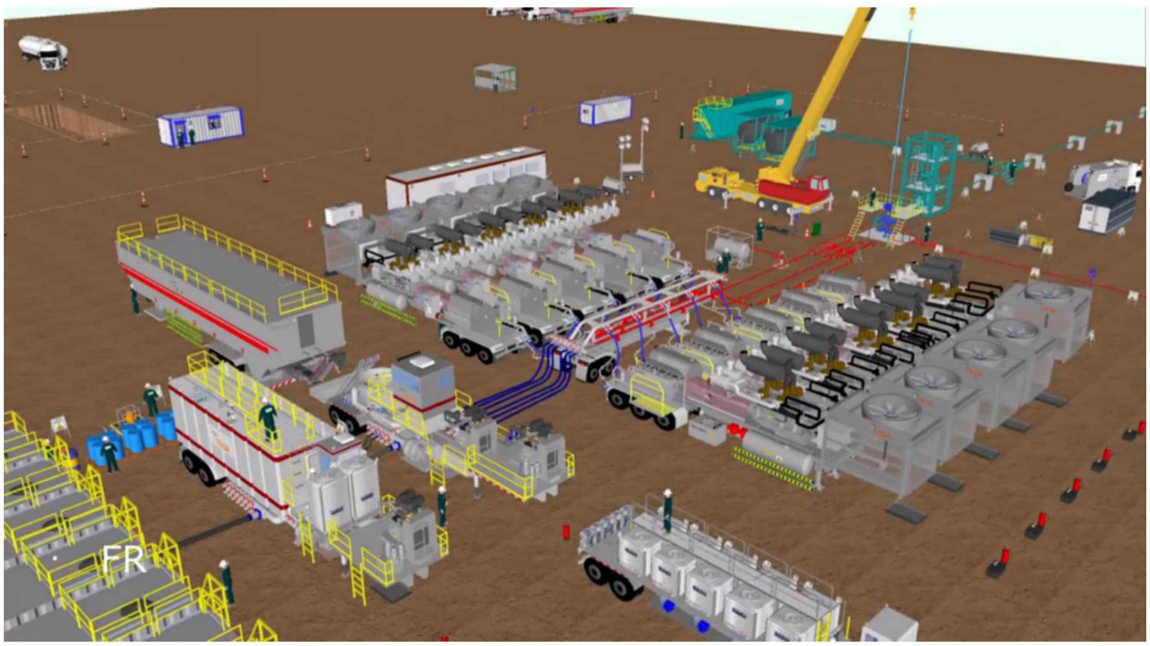


Figure 2—Typical fracturing operation equipment layout demonstrating significant number of resources involved.

Quality Management Systems as a Basis for Reliable Fracturing Operations

The service company's Integrated Management System (Quality & HSE) combines all business components into a coherent system, addressing different objectives but overlapping in management activities to achieve overall organizational goals. The framework includes nine elements of the continuous improvement model: Plan, Do, Check, Act (see Fig. 3 for details).

HSE mission is set to ensure that health, safety, security and environmental considerations remain top priorities for management and for all employees. Prevention of accidental risk and loss from process failure becomes a recognized, integral part of our continuous improvement culture. QHSE Management System defines the principles by which frac department conducts operations with regards to health, safety, and the environment. All observations, near misses, and incidents are input into digital platform called SCORE, the computerized HSE tracking management system. SCORE houses all related documents and can be accessed by all employees. As part of the commitment to maintain and recognize a strong and health Safety Culture, weekly and monthly meetings are held with all levels of the organization to review and publicly recognize and reward the individuals with the best and most QHSE submissions.

Figure 3—QHSE Management System Utilizing PDCA cycle.

Equipment Maintenance

A dedicated maintenance facility was commissioned to address recurring operational challenges, minimize unplanned maintenance costs, and ensure the reliability of fracturing service delivery while establishing a scalable model for efficient fracturing equipment management. The new facility has been strategically planned to tackle recurring operational challenges, aiming to decrease unplanned maintenance costs while ensuring the reliability of our fracturing service delivery. Additionally, it sets up an efficient model for fracturing equipment preservation through planned preventive maintenance and breakdown maintenance. The facility is covering a total area of 11,250 sqm (see Fig. 4).

Figure 4—Fracturing operational base and maintenance facility.

Systematic Preventive Maintenance Program (CMMS) includes thorough multi-level checks. Pre-job and post-job inspections are conducted at the base or well-site to ensure equipment is fit for use and ready for its next job, documenting any faults encountered. Periodic inspections and maintenance are performed every 400-800 operating hours, 180 days, or 10,000 kilometers, inspecting mechanical and electronic components. Repairs and maintenance are recorded using the service company's CMMS platform, accessible via mobile apps. CMMS provides reports on usage history, repairs, modifications, inspections, and test activities, verifying equipment reuse. Maintenance tasks are distributed across operating bases and third parties, utilizing dedicated mechanics and third-party services to reduce maintenance days and increase operating days according to the Preventative Maintenance, Inspection, and Test Program (PMITP).

Utilizing a systematic approach comprising design, implementation, and evaluation to establish a comprehensive maintenance workflow. Customizing maintenance facility designed with ergonomic layouts, advanced diagnostic tools, equipment-specific tool, and maintenance protocols specifically developed for fracturing equipment's. The maintenance protocol established by integrating predictive maintenance technologies, digitalized maintenance management systems, and personnel training programs to ensure process reliability. Validation achieved by conducting a comparative analysis of maintenance process, equipment uptime, and fracturing performance metrics before and after facility implementation and established maintenance protocol.

The implementation of the purpose-built frac equipment maintenance facility has significantly enhanced the reliability and operational service delivery. The implementation of predictive maintenance preventive strategies for equipment preservation minimizing equipment downtime and averting unexpected equipment operation failures. Digitalization of maintenance management system enhances visibility into equipment health, enabling prioritizing critical repairs and minimize operational interruptions effectively. The improved training programs for personnel have resulted in fewer maintenance errors and greater compliance with established procedures.

Since 2022, with an average of two hundred fracturing operation stages totaling 17,007 working hours per year, and only 187.5 hours or 0.1% of Non-Productive Time (NPT) recorded contributed from equipment failures. This performance led to recognition from the client for providing high-quality and efficient service. Furthermore, economic analysis revealed a reduction in unplanned maintenance costs, attributed to streamlined processes and efficient resource allocation. The performance metrics have demonstrated an increase in total productivity of fracturing operations, attributed to the enhanced reliability of frac equipment. The optimized design of the maintenance facility has facilitated faster turnaround times for repairs and component replacements. The tailored frac maintenance facility and protocols enhances equipment maintenance management, reducing incidents and boosting operational efficiency and productivity.

Personnel Competency

The personnel development program called "TAFAWOK" was launched for multiple business lines (CTS, CMT, FRAC, R&M, and other). This program governs the specific career path for every job role for field and operations support employees. Today, those employees will have a path to follow, courses to attend, practical assessments to complete, and lead the future of young talented personnel performing at a high safe technical level of delivery.

Along with TAFAWOK, the competency management system has also been launched for profiling the level of competency for all field and operations support personnel. The system allows for accurate selection of individuals for the job. It was created by first, self-assessing everyone, followed by line managers' validation, and lastly creating a score system online that can be easily accessed to determine the PCP (personnel competency profile) for those members. This plan ensures competency gap closure and creating fully competent personnel in the field.

Focus on training has been prioritized as one of the leading KPI's for company's development. The training and development center (see Fig. 5) has already delivered significant amount of training manhours including technical courses - 25,200 hrs and QHSE - 46,619 hrs as of the end of 2024.

Figure 5—Personnel competency center

Product Development Cycle and Partnerships

Fig. 6 illustrates a typical workflow for technical solution development, highlighting the pivotal role of simulators (Yudin, Nurlybayev et al. 2024). As shown, two layers of modeling are utilized: precise simulators (Planar 3D and Full-3D) for critical wells with sufficient reservoir and fracture measurement data for calibration, and fast, flexible models (P3D) for sensitivity studies and daily operational needs. These models are also more reliable for pressure matching, datafrac analyses, and redesign optimization workflows. An important step is to correlate the daily routine in P3D models with precise models based on representative and well-studied cases with advanced measurements.

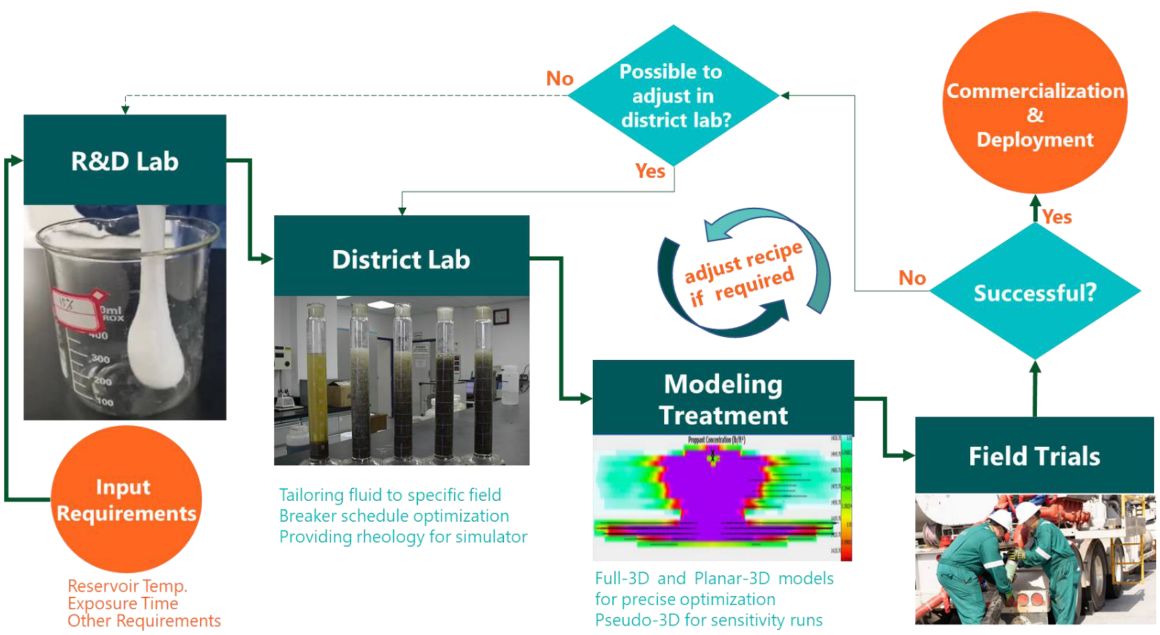


Figure 6—Role of the fracture modeling in the product development cycle.

Developing a tailored solution to dynamically divert fracture within horizontal wellbore can be a good example of successful partnerships. Advantage of start-up in national service companies is flexibility of choosing and creating the partnerships when compared to large international service companies who are typically not eager to take third party solution into their portfolio. Dissolvable diverter co-developed with one of the leading US-based technology and manufacturing companies allowed to find a tailored solution for conventional fracturing applications in KSA. Fig. 7 demonstrates the yard test performed specifically for the condition of KSA operations. Such yard test is the most accurate simulation of the downhole geometries and therefore helping to select the best product composition as a result of yard test and simulation in specialized software.

Figure 7—Yard test to optimize degradable diverter composition and tailor it to middle eastern geological and operational conditions.

Computational fluid dynamics (CFD), lab testing, yard testing, and field trials have been crucial in identifying a framework for polymer selection that adheres to geological and operational conditions. These methods have demonstrated how diverter properties such as crystallinity, amorphousness, degradability, specific gravity, size and shape play a vital role in diverter performance. Selecting a polymer with the proper characteristics is essential for effectively transporting particulates to the desired zone and forming a robust plug during the fracturing treatment. Furthermore, Extensive field trials suggest that improper polymer selection may have adverse effects on production.

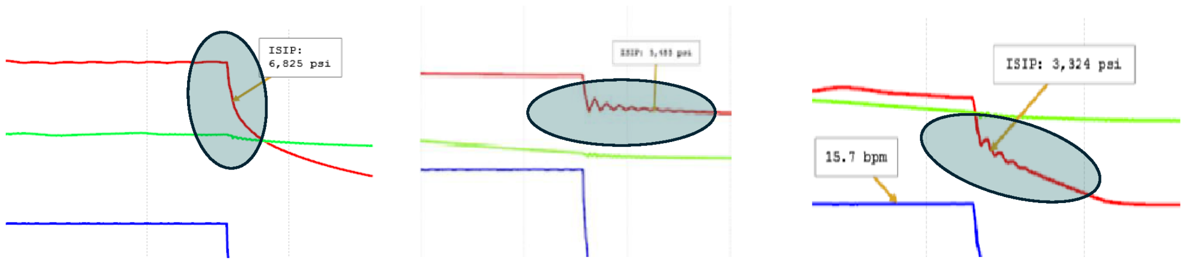
The challenge of the middle eastern application was to match the need from reservoir in large-size particles with operational consideration of material pumpability through surface frac equipment. Conventionally, wide fractures in conventional reservoirs were attempted to plug with large-size spherical shape particulates, that were damaging to pump fluid ends and therefore risky for operational practices. As a result of extended laboratory-scale and field-scale testing it was shown that large-size flake-shape particulates accompanied by smaller sizes granular plugging mixture. All materials were made from fully dissolvable materials capable to withstand temperatures up to 350 F for several hours in such way that the diverting mix is reliable for the fracturing operations and non-damaging to reservoir and wellbore afterwards. Materials screening and selection was made at the lab scale initially using existing industry-wide correlations and then tested at reservoir-scale yard test facility based on the range of temperature and widths from downhole geological conditions and fracture design outputs. Differential pressure of 1000 psi and plug stability for several hours was consistently achieved showcasing a reliable approach to optimize the diverting mixture to operational and geological conditions.

Case study. Local best practices for proppant fracturing in clastic reservoirs

This section introduces a comprehensive workflow designed to enhance the calibration of friction pressures, thereby improving decision-making processes in critical fracturing operations. By integrating advanced methodologies with region-specific adaptations, this workflow addresses the challenges and variabilities inherent in these HPHT reservoirs. The proposed approach aims to refine current calibration practices, leveraging both theoretical and practical insights to achieve more reliable and accurate pressure measurements.

We delve into the intricacies of the calibration process, offering a detailed examination of the workflow's components and their impact on operational outcomes. Case studies and practical examples underscore the effectiveness of this approach, demonstrating how enhanced calibration can lead to optimized fracturing results and better-informed decision-making. This article aims to provide industry professionals with a robust framework for friction pressure calibration that not only meets the demands of today's fracturing operations but also sets a new standard for excellence in the field. Over 180 proppant frac operations were performed using different fluid systems and types of proppants.

Since the fracturing campaign was conducted without direct bottom hole pressure measurements, it was critical to collect only representative wells and treatments for the database to be used as the field input for the friction coefficients. Fig. 8 distinguishes various scenarios of the diagnostic pressure-rate-concentration (PRC) charts. The criteria for selecting a well candidate to be representative for the database are as follows:



a) Bad hammer b) Good hammer and normal leak-off c) Good hammer but high leak-off

Figure 8—PRC plot snapshots with different types of hammers and leak-off.

A minimum of five pronounced oscillations after pumps shut in (the "water hammer effect" or "hammer"). For the purposes of this paper, we will refer to pronounced oscillations as a "good hammer" and less pronounced ones as a "bad hammer." This makes physical sense when correlating the water hammer quality with near-wellbore challenges, as described by Nguyen et al. (2021) and Alowaid et al. (2023).

- Pronounced water hammer indicates good connectivity between the fracture and the wellbore, which means low friction pressure losses in the near-wellbore zone. Therefore, this is a main criterion used for our study to eliminate uncertainties and focus only on wells with friction losses along the tubulars, with minimal losses in the near-wellbore.
- Normal leak-off behavior. This criterion is secondary, but often due to severe leak-off, it was not possible to precisely extrapolate the pressure line and define the pressure at shut-in.

The second chart (b) meets these criteria and is therefore a suitable candidate. In contrast, the first and third charts (a) and © are unsuitable due to the inability to accurately determine the Initial Shut-In Pressure (ISIP).

Such a consistent and accurate approach allowed the field engineers to input the correct fluid coefficients into simulating software to be run during the treatment. Fig 9 provides a workflow that was used by engineers. Two criteria identified to select a representative candidate for the database – clear hammer effect and sufficient fluid efficiency.

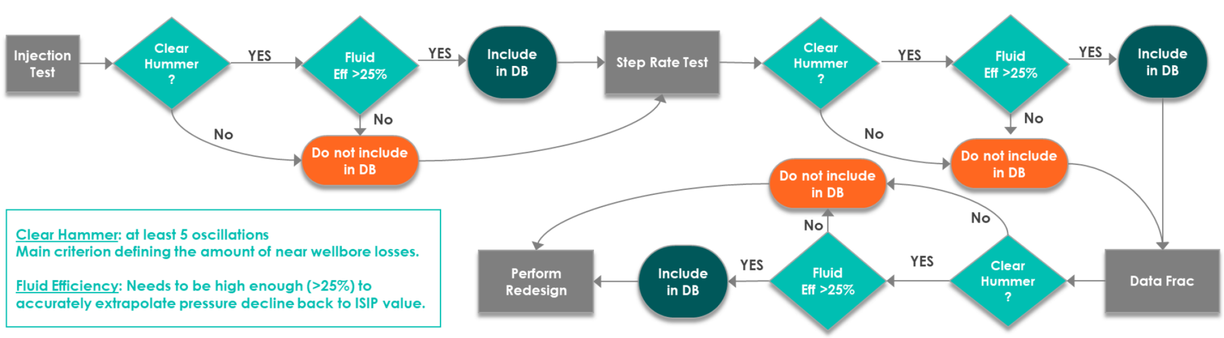


Figure 9—Engineering workflow to build consistent and representative database of fluid friction gradients.

Fig. 10 illustrates the process of data collection and filtering, showcasing several representative wells selected from injectivity tests. The injectivity test is the first to be pumped and is therefore crucial for the engineering team to assess risks and uncertainties associated with near-wellbore (NWB) issues. Special attention is required to define the fluid properties and its friction gradient. "Treated water" is consistently used for these purposes by fracturing fleets, with a regular pumping rate of around 15 barrels per minute (bpm) to establish a representative workflow and decision-making capabilities.

Tubing 3.92 ID lenth, ft	Liner 3.82 ID lenth, ft	Tubing /liner ratio	Rate, bpm	Total frictions, psi	Final treated water gradient	FG/rate psi/1000ft/bpm	standard deviation	Coefficient of variation
7482	7,848	**1.0**	14.0	1,540	100	7.2	0.21	
7482	7,658	**1.0**	15.0	1,448	96	6.4	0.12	
7482	7,513	**1.0**	15.0	1,531	102	6.8	0.01	
7482	7,213	**1.0**	15.0	1,499	102	6.8	0.01	
9019	5,931	**1.5**	15.0	1,476	99	6.6	0.02	
5000	8,815	**0.6**	15.0	1,501	109	7.2	0.28	
5000	8,605	**0.6**	15.0	1,405	103	6.9	0.03	
5000	8,505	**0.6**	15.0	1,346	100	6.6	0.01	
5000	8,220	**0.6**	15.4	1,280	97	6.3	0.18	
8976	5,110	**1.8**	15.2	1,362	97	6.4	0.13	
Average parameters						**6.7**	**0.31**	**4.7%**

Figure 10—Data collection and filtering to reach trustworthy statistical results.

The table in the figure displays the completion lengths of the tubulars and liners for each well in various proportions. Although tubulars and liners have slightly different internal diameters, the plot on the right side of the figure shows that all have similar friction within the measurement error range. At this point, there is no need to distinguish friction gradients (FG) separately for each type. However, acquiring a larger number of representative wells will allow for more accurate allocation of friction gradients for tubing and liners separately. It is critical to filter out wells with FG outliers from the statistics to identify only representative datasets for decision-making. As shown, all wells in this injectivity testing dataset fall within a 5% variation coefficient, indicating a reliable statistical dataset despite the limited number of wells used for the initial estimate.

Fig. 11 presents an extensive FG dataset derived from broader pumping tests, including step rate tests (SRT) conducted with treated water at various rates. The plots indicate a high correlation coefficient (R^2 very close to 1.0), suggesting that the statistical analysis is reliable and can be used for precise evaluations. The left plot of the figure shows a linear relationship between FG and pumping rate, which is convenient for field operations. The right side of the figure provides a classical log-log plot, demonstrating a consistent and direct dependence of FG on pumping rate across a wide range of tests, with rates varying from 14 bpm to 45 bpm.

As a summary for this section, it was shown how to minimize uncertainty and maximize accuracy of the decision making process regarding pressure values in HPHT proppant fracturing operation without the need to use expensive bottom hole gauges. A workflow is presented how to select a representative candidates to serve as a basis for the friction gradient curves used later in decision making process.

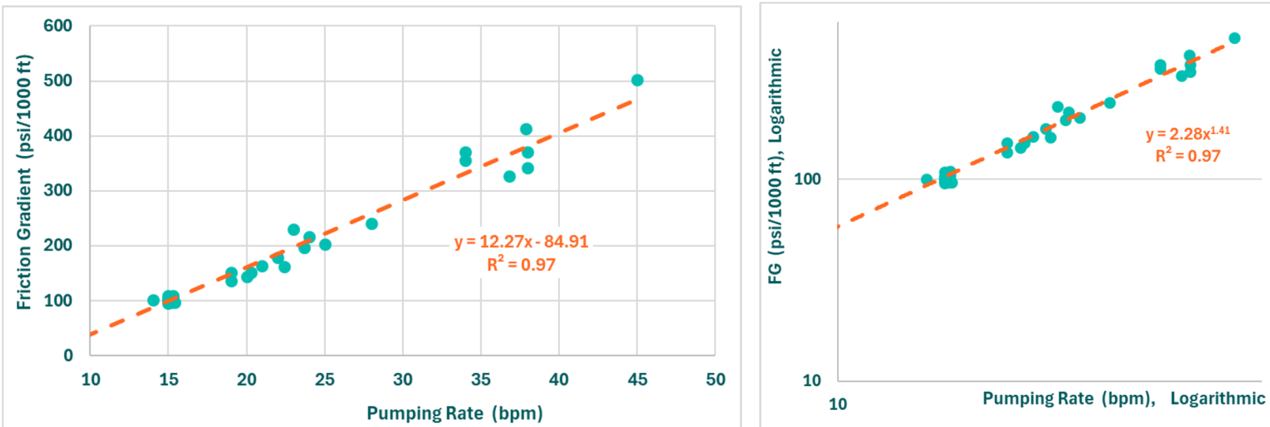


Figure 11—Overall statistics with treated water friction gradient at various pumping rates. Linear plot (left) and logarithmic plot (right).

Case Study. CO2 foam fracturing in carbonate reservoirs

Foamed CO_2 treatments provide effective solutions for hydrocarbon extraction in reservoirs experiencing significant pressure depletion for several decades already (Briscoe. 1979, Ford 1981). In the Middle East, where subterranean carbonate formations are prolific, foamed CO_2 acidizing has been used to target deep, high-temperature, sub-hydrostatic conditions within these formations. A key advantage of this method is its ability to eliminate the need for post-treatment nitrogen lifting via coiled tubing intervention to initiate flowback. In conventional acidizing, nitrogen lifting is often required to initiate well kickoff in depleted reservoirs. The energized nature of the foam, primarily due to the internal phase of CO_2, provides lifting energy during flowback through foam expansion once fracture pressure is released. Moreover, foamed acidizing enhances treatment effectiveness by improving acid placement (due to better fluid loss control and increased viscosity), increasing acid efficiency, and assisting formation cleanup.

The equipment requirements for foamed CO_2 acid fracturing are similar to those of conventional acid fracturing treatments, with the key difference being the additional of specialized equipment to safely store, handle, and inject CO_2 in its liquid phase. Safety precautions and careful operational controls are required to manage and reduce the potential risks of handling high-pressure liquid CO_2, at the surface of the wellsite. Fig. 12 shows a typical layout at the wellsite. The equipment is split into two main areas: one for the standard acid fracturing setup, and another dedicated to CO_2 handling. A separate high-pressure line, dedicated to pumping CO_2, downhole is rigged up and connected directly to the wellhead, allowing better control and monitor of CO_2. Mixing of liquid CO_2, and other fluids (acid, gel, and diverter) occurs at the wellhead.

Once transferred from its transport bulker, liquid CO_2, is stored in pressure vessels at 100 psi or up, depending on weather conditions. Each vessel is equipped with its own pressure-relief valve to safely discharge CO_2, to atmosphere when required, releasing pressure buildup in the vessel. These vessels are then connected via CO_2,-rated hoses to a boost centrifugal pump that will increase pressure when feeding liquid CO_2, to high-pressure fracturing pumps (HP pump). Suction manifold of the HP pump does not utilize any vitaulic or clamp connection, which could cause CO_2 leakage. Every fluid end of the pump is configured to have its own CO_2 venting setup (a bleed-off assembly with choke to discharge it to the atmosphere) operated by remote actuated valve as shown in Figure 3. In the discharge line of the fluid end, isolation and dart check valves are installed before lateral connecting to the main CO_2 high pressure line. In the high-pressure line, additional devices are installed to control pressure and ensure safe operation: venting valve, bleed-off assembly, dart check valve and isolation valve.

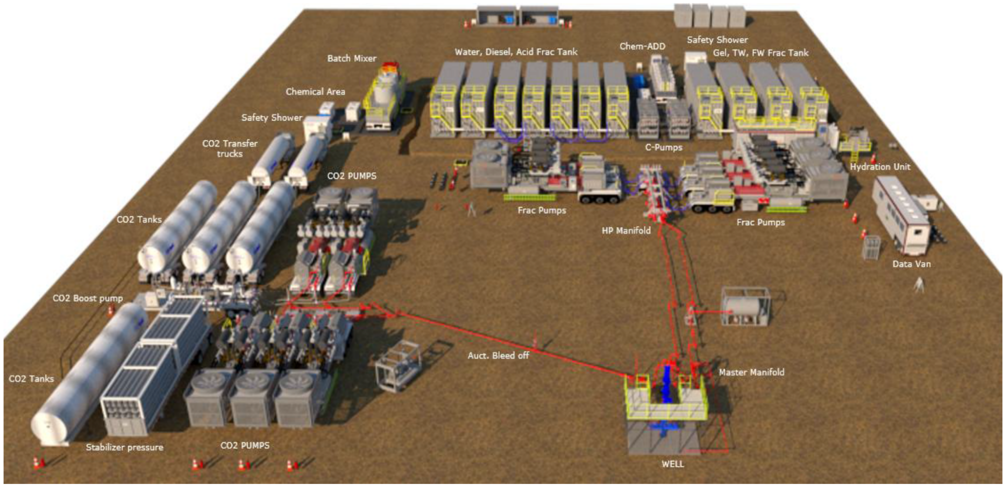


Figure 12—Typical equipment layout for foamed CO_2, acid fracturing treatment.

This case study highlights the successful application of foamed CO_2 acid fracturing in a carbonate formation. Well A was drilled horizontally to a total measured depth of 17,901 ft, targeting a large, laterally extensive zone. The well was completed with a 4.5-inch × 13.5 lb/ft tubing, a 4.5-inch × 7-inch liner hanger, and lower completion containing multistage fracturing sleeves activated via ball drop in a cemented liner. The lateral length was 4,000 ft, with reservoir pressure estimated at about 4,500 psi and bottomhole temperature around 300°F.

Six fracturing stages were designed to maximize gas extraction, with strategic placement of frac ports in zones having at least 5% porosity. Each stage was designed with acid volume of 300 to 500 gallons per foot of pay zone, maintaining CO_2, foam quality of 60%. The treatment design employed a cyclic sequence of a crosslinked gel (PAD), 26% HCl acid, and diverting agents (both chemical and solid diverters), similar to conventional treatments but with enhanced operational control for CO_2 handling. Unlike conventional acid job, emulsified acid (a mixture of acid and diesel) is not utilized in the foamed acid treatment because it loses stability upon contact with CO_2. Table 1 shows pumping schedule for one of the stages in Well A.

Table 1—Treatment schedule for one of the stages in Well A.

Step #	Step Name	Fluid Name	Fluid Volume	Fluid Rate	CO2 Rate	Total Pump Rate	CO2 Volume	BH Foam Quality	Pump Time	Total Stage Volume
			(bbl)	(bbl/min)	(bbl/min)	(bbl/min)	(Tonne)	(%)	(min)	(gal)
1	Cool Down Stage	Treated Water	75	8.0	0.0	8.0	0.0	0	9.4	3,150
2	Acid-0	CSA HCl 26%	48	12.0	0.0	12.0	0.0	0	4.0	2,000
3	*Acid Displcmnt	Treated Water	190	12.0	0.0	12.0	0.0	0	15.9	8,000
4	Pad-1	40# LpH Sirocco (CO2)	29	6.0	8.7	14.7	6.7	60	4.8	2,939
5	Acid-1	CSA HCl 26% (CO2)	38	6.0	8.7	14.7	9.0	60	6.3	3,919
6	Diversion-1	Lo-Gard (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
7	Pad-2	40# LpH Sirocco (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
8	Acid-2	CSA HCl 26% (CO2)	52	8.0	11.6	19.6	12.3	60	6.5	5,389
9	Diversion-2	Lo-Gard (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
10	Pad-3	40# LpH Sirocco (CO2)	33	10.8	15.7	26.5	7.8	60	3.1	3,429
11	Acid-3	CSA HCl 26% (CO2)	57	10.8	15.7	26.5	13.4	60	5.3	5,879
12	Spacer	Linear Gel	4	8.0	0.0	8.0	0.0	0	0.5	168
13	Diversion-3	NWB Diverter	4	8.0	0.0	8.0	0.0	0	0.5	168
14	Spacer	Linear Gel	8	8.0	0.0	8.0	0.0	0	1.0	336
15	Pad-4	40# LpH Sirocco (CO2)	38	12.0	17.4	29.4	9.0	60	3.2	3,919
16	Acid-4	CSA HCl 26% (CO2)	62	12.0	17.4	29.4	14.5	60	5.2	6,369
17	Diversion-4	Lo-Gard (CO2)	29	12.0	17.4	29.4	6.7	60	2.4	2,939
18	Pad-5	40# LpH Sirocco (CO2)	43	12.0	17.4	29.4	10.1	60	3.6	4,409
19	Acid-5	CSA HCl 26% (CO2)	62	12.0	17.4	29.4	14.5	60	5.2	6,369
20	Diversion-5	Lo-Gard (CO2)	29	12.0	17.4	29.4	6.7	60	2.4	2,939
21	Pad-6	40# LpH Sirocco (CO2)	43	14.0	20.3	34.3	10.1	60	3.1	4,409
22	Acid-6	CSA HCl 26% (CO2)	62	14.0	20.3	34.3	14.5	60	4.4	6,369
23	Over Flush (CO2)	Treated Water (CO2)	50	12.0	17.4	29.4	11.7	60	4.2	5,144
24	Flush (CO2)	Treated Water (CO2)	67	12.0	17.4	29.4	15.7	60	5.6	6,859
25	Flush no CO2	Treated Water	12	30.0	0.0	30.0	0.0	0	0.4	500

In every stage, prior to the main treatment, injection test was conducted to verity well's injectivity. Once the injectivity of the stage was confirmed, the preparation of materials commenced. This involved transferring raw acid into storage tanks, where it was mixed with the necessary additives. Additionally, gel and treated water were mixed, and liquid CO_2 was transferred from its transportation vessels. With all required resources arriving promptly at the wellsite, and without any delays, the time required between the execution of the main stages was approximately 24 hours. Figure 13 presents the chart illustrating the main treatment process across the stages. Throughout the 6 stages of foamed CO2 treatment of Well A, the following materials were pumped in these quantities: 101,000 gallons of mixed acid, 72,000 gallons of crosslinked gel, 25,000 gallons of chemical diverter, 4,300 pounds of solid diverter, and 1,300 tons of liquid CO2 (covering both cooling down and priming up).

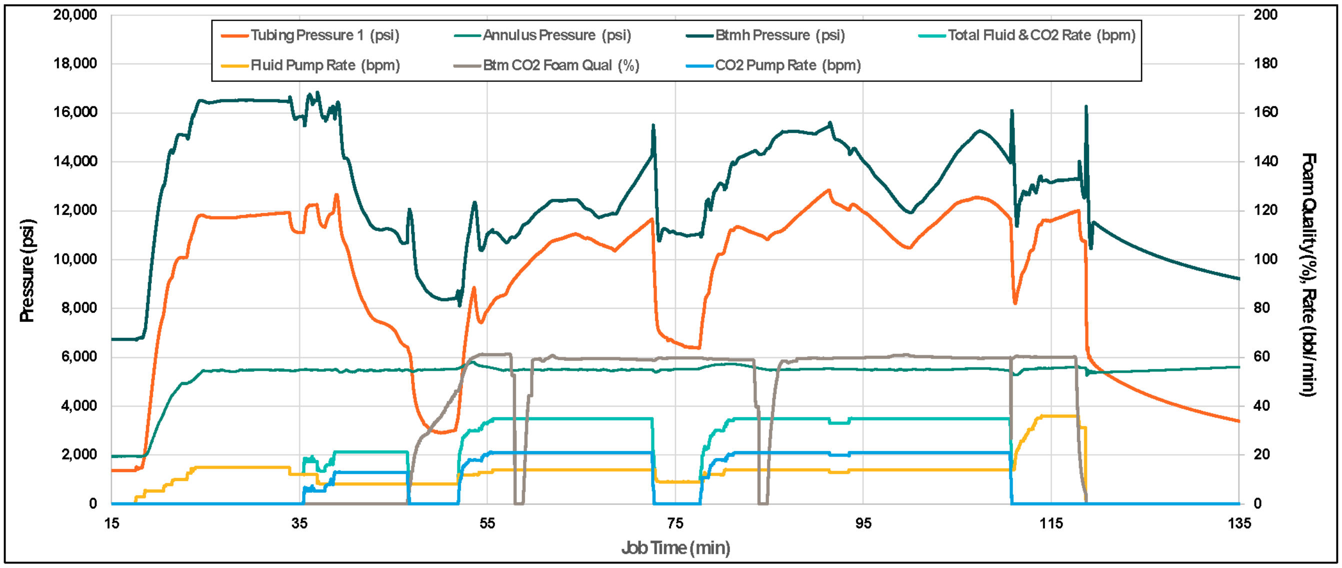


Figure 13—Treatment chart for one of the foamed acid fracturing stages in Well A.

Fig. 14 provides comparison between the fracturing techniques – regular acid fracturing, nitrified acid fracturing and CO2 -based foam fracturing. Gas productivity index (PI) was calculated from initial flowback taking consistent measurement of the rate and pressure drawdown. One can observe a significant advantage of the productivity when CO_2 foam fracturing is implemented. The reason for it is not only additional energy CO_2 stores to improve flowback, but also because the total volume injected into formation was considerably higher in these well. Acid amount was comparable between each acid fracturing operations including nitrified and carbon dioxide foam fracturing, meaning the rock dissolution capacity was comparable. However, when we take into account the gaseous phase – it represents significant amount, especially in case of CO_2 fracturing. Because foam quality of CO2 was 55-60%, meaning that the overall slurry injected into formation was more than doubled when compared to conventional fracturing stages. Even though the acid volume was the amount of rock dissolved was comparable, the length of the fracture propagation was substantially higher in case of CO_2 operations which brings significant benefit for the tight reservoir which was the case in our study. Nitrified fluids had 10-15% foam quality, which places them closer to conventional operations when it comes to generated fracture geometry when compared to CO_2 fracturing.

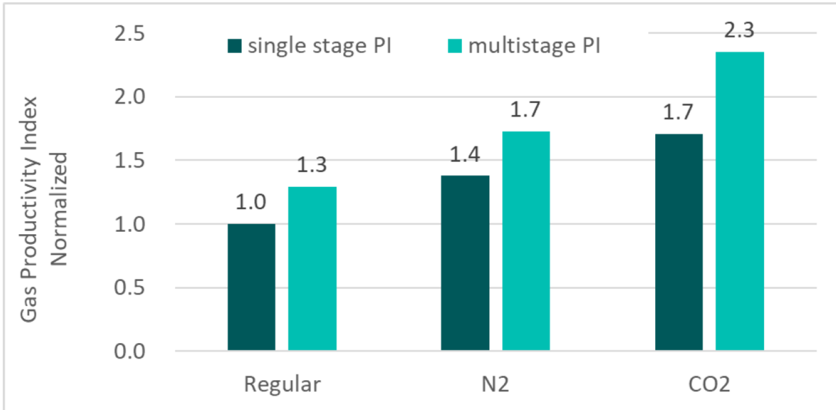


Figure 14—Normalized gas productivity index comparison between regular, nitrified and CO2-based foam fracturing for both single stage and multiple stage well completions

Conclusions

This study provides an overview of the successful project start up of the fracturing services company showcasing the progress from initial stage to an independent hi-tier services provider within a three years timeframe. Key components to reach this sustainable progress where shown to be as following:

- Fundamental and consistent approach to service quality having management system in place and under continuous improvement principles
- Dedicated framework of personnel competency development
- Maintenance program and digitalized platform to ensure flawless exection
- Flexibility to select and partner with know-how industry leaders providers
- Developing local practices and unique product solutions and implementing them in the field

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References

Alowaid, A., Altayyar, T., Mechkak, K. 2023. Water Hammering Evaluation in Perforation Efficiency for Variable Perforation Designs in Unconventional Horizontal Wells. Paper presented at the SPE International Hydraulic Fracturing Technology Conference and Exhibition, Muscat, Sultanate of Oman, September 2023. Paper Number: SPE-215690-MS. https://doi.org/10.2118/215690-MS

Badurayq, S., Bajusair, A., Aljorephani, S. et al 2024. Optimum Supply Chain Management and its Positive Impact on Hydraulic Fracturing Project Start-Up-A Case Study Review. Paper presented at the ADIPEC, Abu Dhabi, UAE, November 2024. Paper Number: SPE-222333-MS. https://doi.org/10.2118/222333-MS

Nguyen, D., Cramer D., Damielson, T. et al 2021. Practical Applications of Water Hammer Analysis from Hydraulic Fracturing Treatments. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, Virtual, May 2021. Paper Number: SPE-204154-MS. https://doi.org/10.2118/204154-MS

Nurlybayev, N., Z. Al-Jalal, M. Samuel, et al 2023. Addressing Hydraulic Fracturing Performance Challenges in HPHT Well with New Fluid System and Comprehensive Fracturing Study Utilizing Full 3D Simulator. Paper presented at the *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, February 2023. Paper Number: SPE-213494-MS. https://doi.org/10.2118/213494-MS

Nurlyabayev, N., Yudin, A., Bulekbay, A. et al 2024. A Comprehensive Study of the Various Stimulation Fluids Deployed in HPHT Reservoirs. Paper presented at the SPE Caspian Technical Conference and Exhibition, Atyrau, Kazakhstan, November 2024. Paper Number: SPE-223439-MS. https://doi.org/10.2118/223439-MS

Yudin, A., Farouk, M., Al-Jalal, Z. et al 2024. Optimization of Fluid Lab Testing for Conventional Proppant Fracturing Applications in HT Reservoirs. Paper presented at the International Petroleum Technology Conference, Dhahran, Saudi Arabia, February 2024. Paper Number: IPTC-24398-MS. https://doi.org/10.2523/IPTC-24398-MS

Yudin, A., Nurlybayev, N., Al-Jalal, Z. 2023. Solving the Low Injectivity Challenges in Hydraulic Fracturing Tight Gas Reservoirs-Case Histories Review. Paper presented at the *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, February 2023. Paper Number: SPE-213424-MS. https://doi.org/10.2118/213424-MS

Yudin, A., Rivas, J., Farouk, M. et al 2024. Fluid Design for Acid Fracturing and Stimulation in HT Reservoirs Accounting for Cooldown Effect Based on Measurements and Simulation. Paper presented at the SPE Caspian Technical Conference and Exhibition, Atyrau, Kazakhstan, November 2024. Paper Number: SPE-223390-MS. https://doi.org/10.2118/223390-MS

Yudin, A., Dawoud, M., Arshad, W. et al 2024. Addressing the Near-Wellbore Injectivity Limitations while Fracturing Highly Pressurized Formations. Paper presented at the SPE Caspian Technical Conference and Exhibition, November 26-28, 2024. Paper Number: SPE-223474-MS. https://doi.org/10.2118/223474-MS